Pitcher Role Reversal

Linear Combination of the 3 metrics helps identifies strong pitchers

I. **Stuff+**:

Stuff+ looks only at the physical characteristics of a pitch. Important features include, but are not limited to, release point, velocity, vertical and horizontal movement, and spin rate. We will use the 8 most common individual types of pitches that have a Stuff+ rating.

Benchmark - 100 ; Above 100 - Good ; Below 100 - Bad

II. **xERA**:

Expected ERA (xERA) depicts a pitcher's "True ERA" as it bases its expectation on factors within a pitcher's control. These factors include Hits, Walks, Home Runs allowed and K's. When you consider the inequity with the ERA calculation, specifically related to errors and base runners inherited/left, xERA is considered to be a more precise gauge of ability.

III. Location+:

“Location+ is a count- and pitch type-adjusted judge of a pitcher’s ability to put pitches in the right place. No velocity, movement, or any other physical characteristics are included in the statistic.” -fangraphs.com

Classification Methodology

1. Identify the historical (2021-2023) pitch type characteristics of starting pitchers (SP), closers (CL), and non closer relief pitchers (RP) based on Stuff+ pitching model. With a focus on SP and CL, we group these pitchers into 6 groups based on xERA (what they control):
   - Successful SPs (xERA < 4.35), unsuccessful SPs otherwise
   - Successful RPs (xERA < 4.50), unsuccessful RPs otherwise
   - Successful CLs (xERA < 4.00), unsuccessful CLs otherwise

2. To be successful, a starting pitcher will need a variety of above average pitches to be able to be effective at getting batters out and not being predictable (pitch more than 5 or more innings during a game).

3. We evaluated each of the above 3 groupings based on the number of above average pitches (individual pitch Stuff+ rating > 100) each pitcher throws.

We focussed on:

- How many of the 7 most common Stuff+ rated pitches are above average (>100)?
- How many of the 7 most common Stuff+ rated pitchers are in the 95th percentile for that pitch type?

We use this analysis to assess if a pitcher that is unsuccessful in their current role may have the attributes of a successful pitcher in an alternate role.

The Candidates

RP -> SP Candidates

Considerations:

- Definition of an RP: Less than 30% of Games Played (GP) in relief role and less than 5 saves in 2023
- 3+ above average different pitches (>100 Stuff+)
- Currently a "Successful RP" (xERA < 4.50)
- Good control (low BB/9 and high Location+)
- Age (< 30 yr. old): may be injury risk to "stretch" an older pitcher out from usually throwing 30-40 pitches/game to 60+ pitches/game
- Factor in expected pitch velocity decline that is expected from throwing more pitches in a game (lower weighting of the "fastball" pitch type)

Name	Age	Fastball	Sinker	Cutter	Splitter	Slider	Curveball	Changeup	Knuckle Curve
Josh Winckowski	***25***	***100***	***102***	***135***	***--***	***108***	***--***	***104***	
Daniel Palencia	23	142	147	--	--	137	145	75	---
Beau Brieske	25	110	109	--	--	113	--	108	--
Gabe Speier	28	92	103	--	--	110	--	100	--

Closer Candidates

Criteria Considerations:

- Not currently a Successful SP (< 4.35 xERA) and not a current CL (< 5 saves)
- Definition of a CL Candidate: Less than 5 saves in 2023
- 2 above average different pitches (>100 Stuff+)
- At least one "elite" pitch (95th Percentile by Stuff+)
- If current SP, has a significant Stuff+ drop off with 3rd pitch
- Good control (low BB/9 and high Location+)
- Displayed ability of being more effective in a one inning role

Name	Curr Role	Fastball	Sinker	Cutter	Splitter	Slider	Curveball	Changeup	Knuckle Curve
Graham Ashcraft	***SP***	*--*	***56***	***116***	*--*	***165***	*--*	*--*	*--*
James McArthur	RP	63	94	103	--	103	141	--	--
Griffin Jax	RP	110	101	119	--	149	--	96	--
Brock Stewart	RP	117	107	106	--	153	--	91	--

SP -> RP Candidate

Criteria Considerations:

- Definition of an SP: Great than 30% of GP as a starting pitcher
- 2 above average different pitches (>100 Stuff+)
  - Or one elite pitch could work (not ideal) - but definitely should not be SP
- Significantly below average 3rd pitch (bottom x% pitch by Stuff+)
- Unsuccessful SPs (xERA > 4.35)
- Age: Exclude very young pitchers that are just developing (< 26 years old)

Name	Age	Fastball	Sinker	Cutter	Splitter	Slider	Curveball	Changeup	Knuckle Curve
Ty Blach	33	--	68	170	--	--	66	20	--
Chase Anderson	*36*	*67*	*69*	*107*	*--*	*104*	*75*	*70*	*--*
Matt Manning	26	102	--	--	--	112	102	62	--

Pitcher Role Switches - In Depth Analysis

1. SP to CL Candidate (Ashcraft)

Name	Curr Role	Slider Stuff+	Sinker	Location+	BB/9	Slider	1st Inning ERA	2nd Inning ERA	Fastball Velo
Graham Ashcraft	*SP*	*165*	*56*	*99*	*3.21*	*165*	*2.08*	*7.36*	*96.5 mph*

Split	G	IP	ER	ERA	PA	AB	R	H	2B	3B	HR	SB	CS	BB	SO	SO/W	BA	OBP	SLG	OPS
1st inning	26	26.0	6	2.08	104	90	6	17	3	0	1	4	0	9	20	2.22	.189	.291	.256	.547
2nd inning	26	25.2	21	7.36	121	106	20	38	3	1	5	3	2	12	25	2.08	.359	.430	.547	.977
3rd inning	25	24.2	13	4.74	102	92	13	22	3	0	3	1	1	8	17	2.13	.239	.314	.370	.683

- Considerations:
  - Not currently a Successful SP (< 4.35 xERA)
  - 2 above average different pitches (>100 Stuff+)
  - Elite Slider in the 95th percentile for Stuff+
  - Has a significant Stuff+ drop off with 3rd pitch: Sinker: 56 Stuff+
  - Good control (3.21 BB/9 and 99 Location+)
  - Quality Fastball Velo (96.5 mph)
  - Effective against Right Batters (.287 BA, 12 HR allowed in 327 PA) and Left-handed (.243 BA, 11 HR allowed in 297 PA)
  - Displayed ability of being more effective in a one inning role (2.08 ERA, .189 BA against in 1st Inning over 26 games)

2. RP to SP Candidate (Winckowski)

Name	Age	Sinker	Cutter	Slider	Changeup	xERA	Location+	BB/9	Innings Pitched (2023)
Josh Winckowski	*25*	*102*	*135*	*108*	*104*	*3.71*	*101.9*	*3.31*	*84.1*

Pitch Count Pitch Data is for 1988 on Share & Export ▾ Glossary

Split	G	PA	AB	R	H	2B	3B	HR	SB	CS	BB	SO	SO/W	BA	OBP	SLG	OPS
Pitch 1-25	60	305	279	34	76	14	0	7			23	67	2.91	.272	.334	.398	.732
Pitch 26-50	21	60	53	8	13	1	0	2			7	14	2.00	.245	.333	.377	.711
Pitch 51-75	2	2	1	0	0	0	0	0			1	1	1.00	.000	.500	.000	.500

- Considerations:
  - Definition of an RP: Less than 30% of GP in relief role and less than 5 saves in 2023
  - 4 above average different pitches (>100 Stuff+)
  - Currently a "Successful RP" (xERA = 3.71)
  - Good control; low BB/9 (3.31) and high Location+ (101.9)
  - Age (25 yr. old): may be injury risk to "stretch" an older pitcher out from usually throwing 30-40 pitches/game to 60+ pitches/game
  - A high number of Innings Pitched for a reliever role (84.1) where the transition to a SP may not be as drastic as someone with a low inning count
  - Factor in expected pitch velocity decline that is expected from throwing more pitches in a game (lower weighting of the "fastball" pitch type)
  - Effective against Right Batters (.287BA, 8 HR allowed in 216 PA) and Left-handed (.239 BA, 1 HR allowed in 151 PA)
  - Effective pitcher after throwing 25 pitches (actually improves after 25 pitches)

3. RP to SP Candidate (Jax)

Name	Age	Fastball	Sinker	Cutter	Slider	xERA	Location+	K/9	BB/9	Innings Pitched (2023)
Griffin Jax	*29*	*110*	*101*	*119*	*149 (Elite)*	*2.88*	*105.8*	*9.37*	*2.62*	*65.1*

Platoon Splits Share & Export ▾ Glossary

Split	G	PA	AB	R	H	2B	3B	HR	SB	CS	BB	SO	SO/W	BA	OBP	SLG	OPS	TB	GDP	HBP	SH	SF	IBB	ROE	BAbip	tOPS+	sOPS+
vs RHB	70	157	145	12	33	4	0	3	3	0	8	33	4.13	.228	.282	.317	.599	46	2	3	1	0	0	2	.275	87	66
vs LHB	64	117	105	16	25	5	2	2	6	0	11	35	3.18	.238	.316	.381	.697	40	3	1	0	0	1	1	.338	117	89

Pitch Count Pitch Data is for 1988 on Share & Export ▾ Glossary

Split	G	PA	AB	R	H	2B	3B	HR	SB	CS	BB	SO	SO/W	BA	OBP	SLG	OPS	TB	GDP	HBP	SH	SF	IBB	ROE	BAbip	tOPS+	sOPS+
Pitch 1-25	71	265	243	24	56	9	2	4			17	64	3.76	.231	.292	.333	.625	81	5	4	1	0	1	3	.297	95	73
Pitch 26-50	7	9	7	3	2	0	0	1			2	4	2.00	.286	.444	.714	1.159	5	0	0	0	0	0	0	.500	257	213

- Considerations:
  - 4 Quality Pitches including 1 Elite (Slider)
  - Currently very successful RP (xERA = 2.88)
  - Higher end of IP for a RP → transition to SP not as drastic as a 30 IP player
  - Similar effectiveness against both RHB and LHB (will be able to handle diverse batters)
  - Pitches 26-50 show a slight decrease in production, but small sample size and abnormally high BABIP
    - small sample size
  - High K/9 and low BB/9 stats
  - Good age (middle of prime); age and injury risk not as big a concern

4. Bonus Candidate:
Ty Blach: SP to Left Handed Specialist

Name	Age	Hand	Sinker	Cutter	Curveball	Changeup	LHB BA	xERA	WHIP
Ty Blach	33	L	68	170	66	20	.269	7.21	1.641

Pitch Count Pitch Data is for 1988 on Share & Export ▾ Glossary

Split	G	PA	AB	R	H	2B	3B	HR	SB	CS	BB	SO	SO/W	BA	OBP	SLG	OPS	TB	GDP	HBP	SH	SF	IBB	ROE	BAbip	tOPS+	sOPS+
Pitch 1-25	20	115	108	12	37	7	0	3			6	16	2.67	.343	.374	.491	.865	53	4	0	0	1	0	1	.378	90	137
Pitch 26-50	18	117	104	20	36	6	1	4			8	16	2.00	.346	.393	.539	.932	56	3	2	0	3	0	0	.368	103	155
Pitch 51-75	13	79	66	8	17	0	1	3			8	12	1.50	.258	.354	.424	.779	28	3	3	0	2	0	0	.264	72	110
Pitch 76-100	11	40	36	12	14	3	0	5			2	6	3.00	.389	.425	.889	1.314	32	2	1	0	1	0	0	.346	177	232

Platoon Splits Share & Export ▾ Glossary

Split	G	PA	AB	R	H	2B	3B	HR	SB	CS	BB	SO	SO/W	BA	OBP	SLG	OPS	TB	GDP	HBP	SH	SF	IBB	ROE	BAbip	tOPS+	sOPS+
vs RHB	20	280	248	43	86	14	1	11	1	1	21	40	1.90	.347	.400	.544	.944	135	11	5	0	6	0	0	.370	106	158
vs LHB	20	72	67	9	18	2	1	4	1	0	3	10	3.33	.269	.306	.508	.813	34	1	1	0	1	0	1	.259	75	116
vs RHB as LHP	20	280	248	43	86	14	1	11	1	1	21	40	1.90	.347	.400	.544	.944	135	11	5	0	6	0	0	.370	106	148
vs LHB as LHP	20	72	67	9	18	2	1	4	1	0	3	10	3.33	.269	.306	.508	.813	34	1	1	0	1	0	1	.259	75	132

No real trends as the game progresses. Higher BA's than a SP should have.

Opposing BA gets considerably better against LHB

- Considerations:
  - #1 Stuff+ ranked Cutter ball in the MLB; no other quality pitches
  - Performs better against LHB
  - xERA + WHIP in combination with only 1 Quality Pitch means Blach should not be utilized as a SP
  - His skill set could be harnessed in a more situational role.
  - Too volatile to trust for full innings/multiple innings
    - 1 or 2 LHB max